



Project Site

FlatLands

Generative Floormap Completion from a Single Egocentric View

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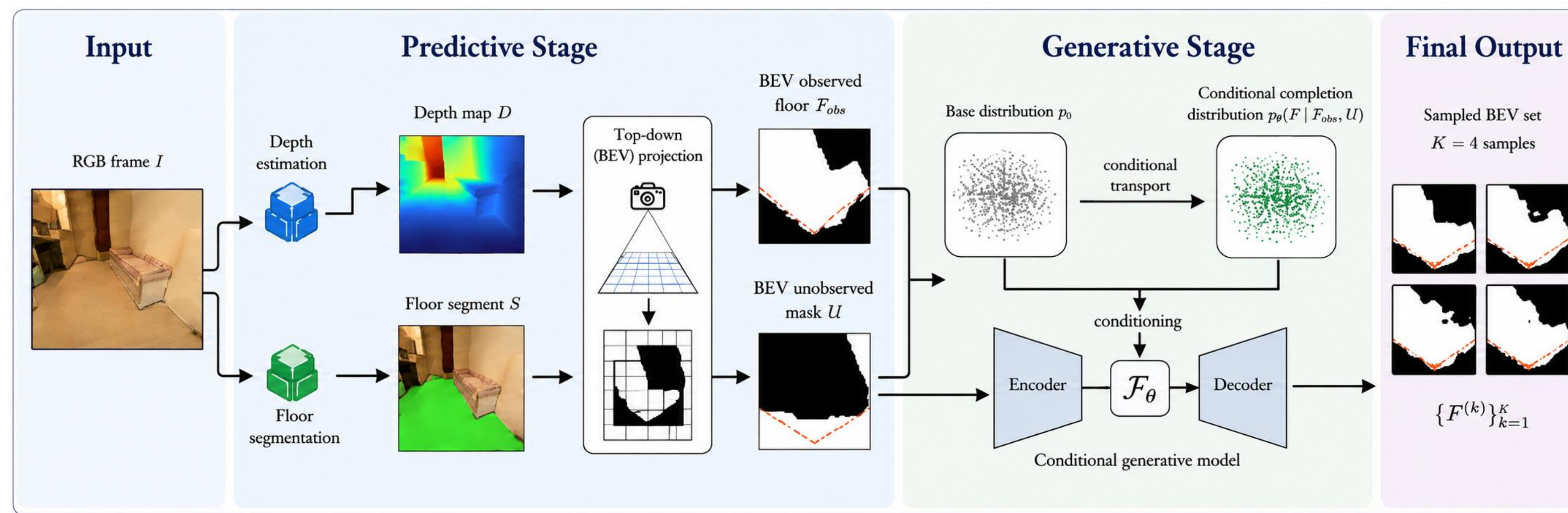


Datasets & Benchmarks
Computer Vision for Robotics

Floormap Completion

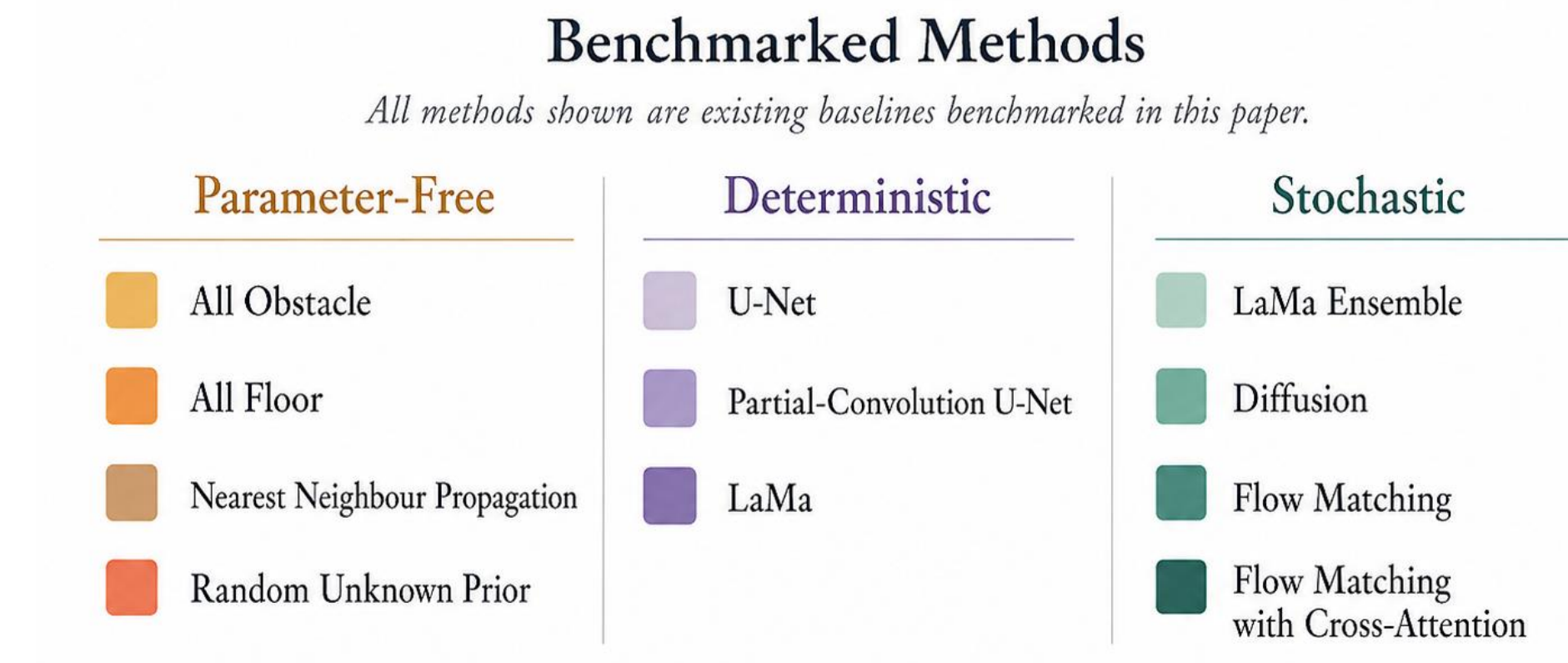
Problem: Predicting a complete bird's-eye-view traversability map from one egocentric RGB image to reason beyond the field of view of the visible scene.

Challenge: Unobserved regions admit multiple plausible layouts, motivating conditional 2D generative completion with deterministic and stochastic baselines.



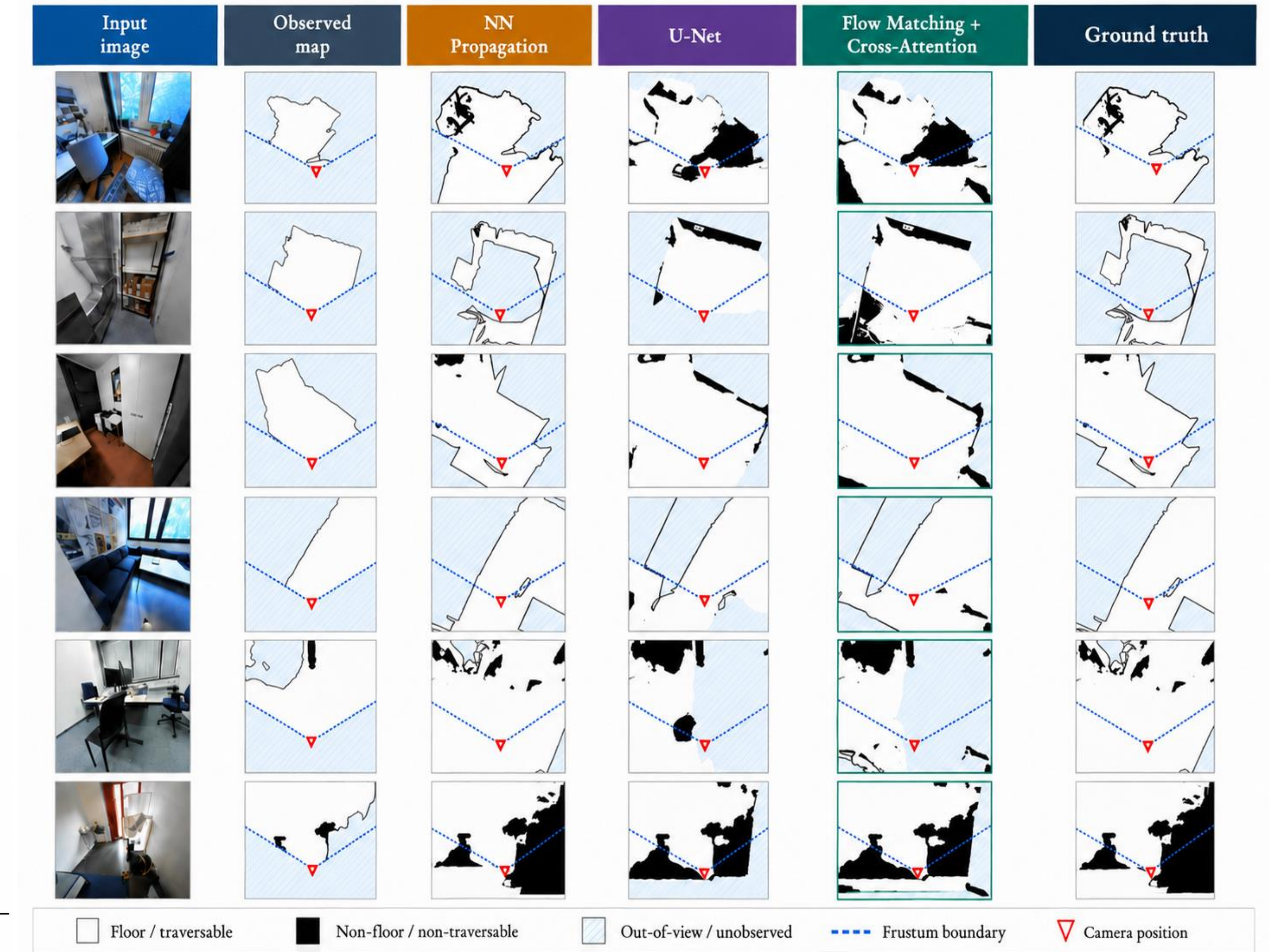
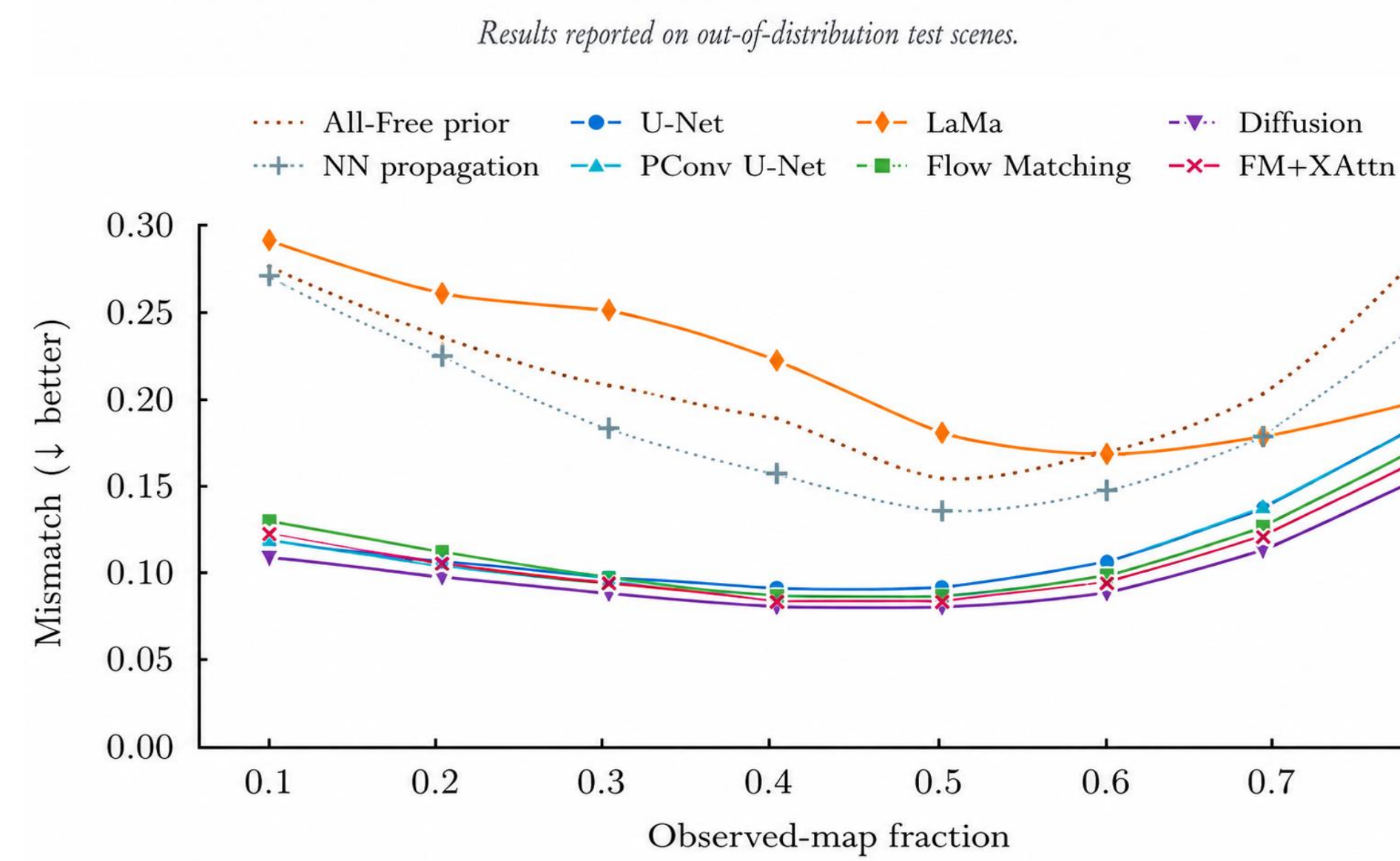
FlatLands systematically benchmarks established deterministic and stochastic methods for floormap completion, rather than proposing a new architecture.

Benchmark & Results



Key Quantitative Findings
Flow Matching with Cross-Attention performs best overall across the benchmark.

- Completion coverage**: Highest coverage of plausible completions among the compared methods.
- Distribution error**: Lowest distribution error in the out-of-distribution benchmark.
- End-to-end mismatch**: Lowest mismatch overall across the evaluated methods.



FlatLands Dataset

Dataset Poses

Virtual camera views are paired with geometry-derived floor maps from the underlying 3D assets.

Virtual Camera Poses

Steps used:

- Back projection
- Rasterization
- Orthographic Projection

270,575
Observations

17,656 Scenes **6** Source datasets

Dataset	Scenes	Observations
ScanNet	1,508	24,763
ZInD	7,026	133,096
ScanNet++	927	16,214
3RScan	1,291	18,216
ARKitScenes	4,803	40,282
Matterport3D	2,101	38,004

Main Findings

- Model rankings remain consistent on unseen datasets, showing strong out-of-distribution generalization
- IoU can reward trivial floor guesses; energy scores better capture fidelity and diversity
- Stochastic models preserve plausible alternatives, but severe occlusion erodes their advantage over deterministic models
- Generative models localize uncertainty at ambiguous boundaries; ensembles spread it indiscriminately across the map.
- With RGB input, perception errors dominate and diminish the gains across stochastic completion variants

References

- Datasets:** ZInD'21 • ARKitScenes'21 • Matterport3D'17 • ScanNet'17 • 3RScan'19. • ScanNet++'23
- Baselines:** U-Net'15 • Partial Conv'18 • LaMa'22
- Generative Methods:** GAN'14 • DDPM'20 • DDIM'21 • Flow Matching'23 • Rectified Flow'23
- Pretrained Models:** Depth Pro'25 • SegFormer'21
- Metrics:** Boundary IoU'21 • Energy Distance'16

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